

Original Article

Iodine-125 episcleral plaque therapy in uveal melanoma

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ABSTRACT

Purpose: To assess the efficacy of iodine-125 (I-125) episcleral plaque therapy in the management of uveal melanoma with regard to local control, survival, globe preservation and visual outcome.

Methods: Between January 1985 and January 1997, 50 patients with the diagnosis of uveal melanoma were treated with I-125 episcleral plaques. The mean initial tumour height was 5.5 mm (range, 2.0–9.5 mm) and basal diameter 9.5 mm (range, 4.0–14.5 mm). I-125 seeds with a mean activity of 1259.1 MBq were used. The mean apical dose was 80.5 Gy (range, 68.0–95.0 Gy). Scleral dose ranged from 225.0 to 940.0 Gy. Pretreatment visual acuity was 6/60 or better in 43 patients.

Results: One patient was lost to follow up and excluded from analysis. The remaining 49 patients had a mean follow up of 39.5 months. There were seven local failures, with a mean duration to tumour progression of 16.7 months. All seven patients were successfully managed with enucleation. Five other enucleations were performed for treatment-related complications. At the time of analysis, 10 patients have died, five of metastatic melanoma. Of the patients with preserved globes, corrected visual acuity was 6/60 or better in 31 patients.

Conclusions: Good local tumour control and survival can be achieved with episcleral plaque therapy. Globe preservation with useful vision was possible in the majority of cases. Our findings are in keeping with other reported series.

Key words: episcleral plaque, iodine-125, radiotherapy, uveal melanoma.

INTRODUCTION

Prior to the introduction of episcleral plaque therapy (EPT) in Western Australia, all patients with the diagnosis of uveal

melanoma underwent enucleation as the primary form of local treatment. Enucleation offers good local control at the sacrifice of remaining vision and the globe of the affected eye.

In the search for preservation of vision with local tumour control, EPT has been employed at our centre since 1985.

The optimal treatment of uveal melanoma with respect to survival remains undefined. It is generally agreed that large melanomas are best treated by enucleation and small ones by globe sparing methods.¹ There is less consensus regarding medium-sized tumours. To date, there are no published randomized trials comparing enucleation with EPT. The Collaborative Ocular Melanoma Study (COMS) group is currently undertaking such a study.² Statistical matching of comparable populations treated with the two techniques have not demonstrated significant survival differences.^{3–5}

We report our experience with iodine-125 (I-125) EPT in 50 patients with uveal melanoma, and here assess its efficacy in the management of uveal melanoma with regard to disease control, globe preservation, visual outcome and survival.

METHODS

Between January 1985 and January 1997, 50 patients with uveal melanoma were treated with I-125 episcleral eye plaques at Royal Perth Hospital. This number represented all cases in Western Australia treated with EPT up until January 1997. Patients were identified through hospital records, and their clinical presentation, treatment, and response data were compiled retrospectively.

Treatment was performed in close collaboration with the Lions Eye Institute and the Eye Clinic at Royal Perth Hospital. All patients were evaluated by one of two senior ophthalmologists to document their disease status and suitability for EPT. Each patient underwent full ophthalmologic assessment with visual acuity testing, direct fundal visualization, fluorescein angiography, fundal photography, and B-scan ocular ultrasonography. The diagnosis was made based on the clinical characteristics and growth pattern of the lesion. Systemic metastasis was excluded by physical examination, liver function test and chest X-ray.

Table 1. Tumour stage of patients

Tumour stage	No. patients
T1 (diameter ≤10 mm and height ≤3 mm)	6
T2 (10 mm < diameter ≤5 mm and 3 mm < height ≤5 mm)	17
T3 (diameter > 15 mm or height > 5 mm)	27
T4 (extraocular extension)	0

n = 50.

Abnormalities detected were followed up with computed tomography of the abdomen and chest.

One patient was lost to follow up and was excluded from the analysis. The remaining 49 patients consisted of 29 men and 30 women. The mean age at diagnosis was 62 years (range, 23–84 years). Visual loss or blurring were the most common clinical presentations. Eight patients had ocular complications related to their tumour at the time of diagnosis. Six patients were observed to have had serous retinal detachment, one had secondary glaucoma due to involvement of the iris and one had vitreous haemorrhage. Visual acuities were recorded using a standard Snellen visual acuity chart. The best corrected pretreatment visual acuity (VA) was ≥ 6/12 (ie. better or equal to 6/12) in 31 patients, < 6/12 but ≥ 6/60 in 12 patients and < 6/60 in six patients. Three patients were known to have had another malignancy, one with treated breast carcinoma, one with a resected cutaneous melanoma and one with a resected Duke's B carcinoma of the colon.

All patients had disease confined to the globe. Twenty-one tumours occurred in the right eye and 28 in the left eye. Twenty-eight tumours were situated in the posterior pole of the eye with seven of these involving the macula, 15 were situated anterior to the equator and six were not recorded. All but two tumours originated in the choroid. These two tumours developed in the ciliary body. The mean tumour height was 5.5 mm (range, 2.0–9.5 mm) and the mean tumour basal diameter 9.5 mm (range, 4.0–14.5 mm). Tumours were categorized from T1 to T4 using the American Joint Committee on Cancer staging system for uveal melanoma,⁶ as outlined in Table 1.

Patients were treated with reusable I-125 seed ophthalmic applicators. The applicator consists of a stainless steel cap which houses a perspex insert; the perspex holds the I-125 seeds. Three applicator sizes were available: 10 mm, 12 mm and 15 mm to treat tumours with each respective maximum diameter. The 12 mm applicator was commissioned and available only after January 1995. Details on dosimetry and fabrication of applicator have previously been published.⁷ All plaques were placed on the scleral surface with the patient under general anaesthesia. Conjunctival limbal peritomy was performed and the tumour margins identified using indirect ophthalmoscopy, and delineated with diathermy marks. A metal template was centred over the tumour base to locate stitches on the sclera for fixation of the loaded plaque. The template position relative to the tumour was then rechecked with indirect

Table 2. Tumour treatment details

Treatment detail	Mean	Range
Dose to apex (Gy)	80.5	68.0–90.0
Activity (MBq)	1358	655–3240
Treatment time (h)	106	62–188
Dose rate (Gy/h)	75	42–129
Tumour margin (mm)	0.97	–1.00–2.75

ophthalmoscopy and scleral indentation. Recti muscles were mobilized as necessary to facilitate placement of the plaque. After the calculated interval for radiation delivery was completed, the plaque was removed.

Plaque size was usually selected to ensure a minimum of 1 mm margin coverage of basal diameter. In the early part of the series, only a 10 mm plaque was in service and three patients were treated with inadequate margins. The dose prescription was aimed to deliver 80 Gy to the apex of the tumour over 3 to 4 days. The tumour apex was defined as the tumour height as determined by B-scan ocular ultrasound plus 1 mm to allow for thickness of the sclera. Treatment details are shown in Table 2.

Follow up was performed by the treating ophthalmologist. Ophthalmologic examination and B-scan ultrasound were routinely conducted to assess tumour response. Fluorescein angiography was used as an adjunct in some cases to assess response and also to document treatment complications. Features indicative of regression or growth were noted. The tumour was considered to have regressed if there was the presence of subretinal fluid reabsorption and a change in the observed characteristics of the tumour, varying from gradual collapse to the development of a flat pigmented chorioretinal scar. Growth in tumour dimensions constituted progression. Patients who did not reside in Perth had their subsequent evaluations at the centre of referral. Details on these patients were obtained by mail or telephone correspondence.

One patient was lost to follow up soon after treatment and was excluded from analysis. The mean period of patient follow up was 39.5 months (range, 2.4–127.1 months).

Local control, disease-free survival and overall survival were estimated using the Kaplan–Meier product limit method. The Cox proportional hazard regression model was used to assess for prognostic factors.

RESULTS

Local control

Satisfactory regression of the tumour was noted in 42 of 49 patients (85.7%). Local control for the purpose of this analysis was defined as the absence of tumour progression. Persistent tumour growth necessitated enucleation of seven eyes at 1, 10, 12, 14, 19, 30 and 31 months. Table 3 shows the clinical variables of this subgroup with local failure.

Table 3. Patient subgroup with local tumour treatment failure

Patient	Tumour height (mm)	Tumour diameter (mm)	Margin coverage (mm)	Dose apex (Gy)
1.	6	8	1	5
2.	7	9	0.5	80
3.	6	13	1	81
4.	5	6	2	82
5.	5	13	1	80
6.	6.5	9	0.5	80
7.	6	10	0	80

$n = 7$.

Table 4. Patient deaths from melanoma

Patient	Tumour height (mm)	Tumour diameter (mm)	Margin coverage (mm)	Dose apex (mm)
1.	7	12	-1**	80
2.	6	12	-1**	84
3.	5	10	1	79
4.*	6	10	0	80
5.	4	9	0.5	68

*This patient had local failure requiring enucleation; ** inadequate margins. $n = 5$.

All seven patients who failed EPT were successfully managed by enucleation. The Kaplan–Meier estimate of local control with EPT at 5 years is 80% (Fig.1).

Survival

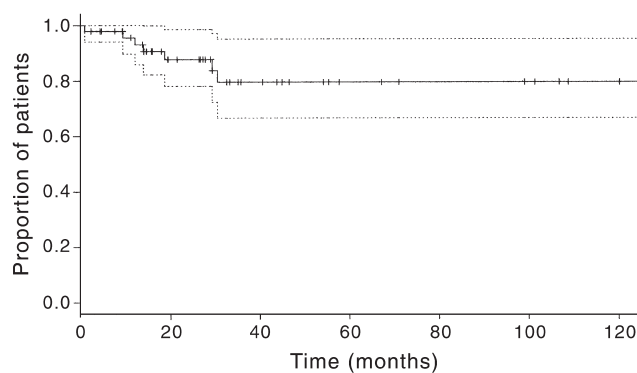
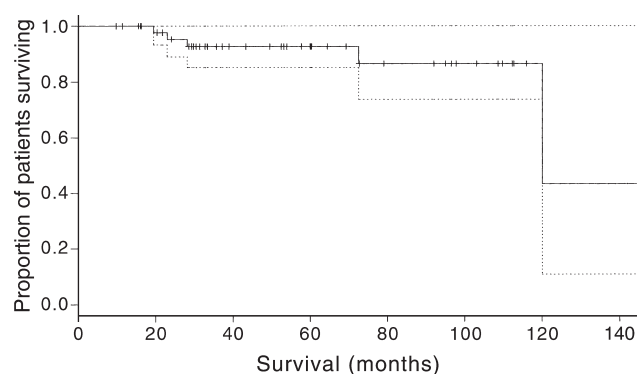
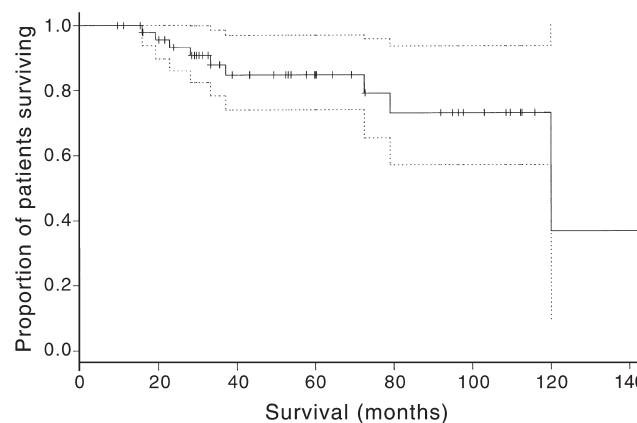
At the time of analysis (1 January 1997), 10 of 49 (20.4%) patients have died. There were five (10.2%) deaths from metastatic melanoma at 23, 29, 73, 80, and 121 months after EPT. Of these melanoma deaths, only one patient had local failure that required enucleation. See Table 4 for subgroup characteristics. The liver was the most common site of metastasis.

The other five deaths were of unrelated causes: one died of acute myeloid leukemia; one of colonic carcinoma; one of breast carcinoma; one of bladder carcinoma; and one of a non-malignant cause.

The estimated 5-year disease-specific survival was 92% (Fig. 2) and overall survival 85% (Fig. 3).

Treatment complications

The most common complications recorded were local radiation retinopathy (20.5%) and retinal detachment (12.2%). Retinal detachments associated with tumour tended to increase in size in the immediate postoperative period before they showed any signs of resolution. Other complications included cataracts, glaucoma, xerophthalmia and optic atrophy.

**Figure 1.** Estimate of local tumour control with episcleral plaque therapy over 5 years. (—), mean; (···), standard deviation.**Figure 2.** Estimated disease-specific survival over 5 years. (—), mean; (···), standard deviation.**Figure 3.** Estimated overall survival over 5 years. (—), mean; (···), standard deviation.

In five patients, enucleations had to be performed for complications causing painful blind eyes: two for massive retinal detachments at 5 and 13 months; two for glaucomas at 29 and 49 months; and one for phthisis bulbi at 59 months.

Table 5. Visual acuity pre and post episcleral plaque therapy

Visual acuity	Pretreatment (n)	Post-treatment (n)
≥ 6/12	31	9
< 6/12 to ≥ 6/60	12	22
< 6/60	6	6
Enucleation	0	12

n = 49.

Visual outcome

Prior to EPT, the majority of patients had good visual acuity (VA) with 31 patients having a VA of ≥ 6/12 in the affected eye. Following EPT, 11 had no change in VA, 37 had deterioration and one had improved VA.

A total of 12 enucleations were performed, seven for tumour progression and five for treatment complications.

Of the remaining 37 eyes: nine had a VA of ≥ 6/12; 22 were in the range < 6/12 to ≥ 6/60; and six were < 6/60 (see Table 5).

In our series, we defined useful vision as having a VA of ≥ 6/60, in other words, vision that allows ambulation or better. Following EPT, 31 patients (63%) were deemed to have useful vision.

Multivariate analysis did not identify tumour height, basal diameter, dose, dose rate, margin of coverage or tumour position as significant predictors for local failure, systemic metastasis or visual outcome.

DISCUSSION

Our results are comparable with other published series. Local control was achieved in 86% of patients with EPT and allowing for enucleations for treatment failures, the local control rate was 100%. One patient had an enucleation performed for tumour progression within a month of EPT. This patient was treated in 1986 when we did not have a full understanding of the latency of tumour response to radiation. In retrospect, enucleation may have been performed prematurely, since early tumour growth may be due to oedema or haemorrhage. Five developed metastases and all eventually died.

The disease-specific survival was 92%. Ocular morbidity was high with more than half the treated patients exhibiting some element of radiation ocular damage, and five requiring enucleations for serious complications. Overall visual outcome was fair with 63% retaining useful vision. Globe preservation was achieved in 76% of patients.

Local control and survival

Uveal melanoma is considered suitable for EPT only if its basal diameter does not exceed 15 mm and apical height is less than 10 mm.⁸ Enucleation should be advised for large tumours due to anticipated morbidity from radiation effects.

The goal of ocular radiation is to destroy the reproductive integrity of tumour cells. Tumour regression is often delayed for many months, with the mean latency of detectable shrinkage being 12 months. Many patients are left with a chorioretinal scar. Objective measurement of response is made difficult by residual pigmentation and radiation scarring. We used fluorescein angiography and ocular ultrasonography as aids to assess disease response.

Accurate delineation of the tumour during plaque application is a challenge. Conventional methods use transillumination and indirect ophthalmoscopy to localize the tumour. Some centres are now using intraoperative ocular ultrasonography to increase the accuracy of plaque placement and thus dose delivery. We plan to investigate the usefulness of this tool in treating future patients.

In terms of tumour coverage, we aimed for a margin of approximately 1 mm, although this did not occur in the majority of cases due to constraints with fixed plaque sizes. It is an interesting observation that two out of the three patients who were treated with inadequate margins (see Table 4) progressed to have metastatic disease, even though their primary site appeared to be controlled. This may reflect the inaccuracy of assessing local tumour control using clinical means. It cannot be determined from our study whether local control had any correlation with deaths from melanoma.

The optimum dose to achieve tumour control and minimize complications for uveal melanoma is undefined despite over 50 years of use. There are many reports of the use of EPT with various isotopes including Co-60, Au-198, Ru-106 and I-125.⁹ I-125 is by far the most popular isotope due to its favourable photon characteristics. Quivey *et al.* reported the results of 150 patients irradiated with 95 Gy to the apex with local control rates of 81% and disease-specific survival of 78% at 5 years with apical doses of 30–141 Gy.¹⁰ Fontanesi *et al.* treated 144 patients with doses ranging from 74 to 84 Gy with local control of 94% and disease-specific survival of 96%.¹¹ Their tumours were mostly medium to large in size. Our results are comparable with these series.

Particle beam therapy has been reported to yield higher local control rates than common brachytherapy techniques. A randomized trial by the University of California San Francisco comparing I-125 EPT to He ions found no survival differences between the two treatment groups, but there were more local failures in those treated with I-125.¹²

Complications and visual outcome

The visual outcome of treated eyes depends mainly on tumour size and location.^{10,13} Larger size tumours are at greater risk of retinopathy and those located close to the optic nerve are at more risk of optic neuritis. Hence, large tumours and those located near the optic disc had the worst visual outcome.

Our series consisted of mostly large T3 tumours with many located in the posterior pole, some with direct involvement of the macula. We hence anticipated substantial ocular

morbidity. In patients who had poor VA prior to treatment, the use of EPT was mainly to preserve the globe, especially in those who refused enucleation. Overall, 63% of our patients retained useful vision of 6/60 or better. This compares favourably with the series by Quivey *et al.*¹⁰ who attained useful vision in 58% and Fontanesi *et al.*¹¹ who reported useful vision retention in 43%.

Our small study sample limited our ability to identify factors which may predict for local failure, melanoma deaths and poor visual outcome on statistical analyses. As the dose delivered to the apex in our series was relatively uniform and the tumour dimensions mainly large, we were not able to demonstrate with statistical tests that dose and tumour size were prognostic factors.

In summary, our data support the view that I-125 EPT is a safe and effective method to treat uveal melanomas. It allows preservation of globe and useful vision in many cases. Survival with this technique appears good, although randomized data are required for a true comparison with enucleation.

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